

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

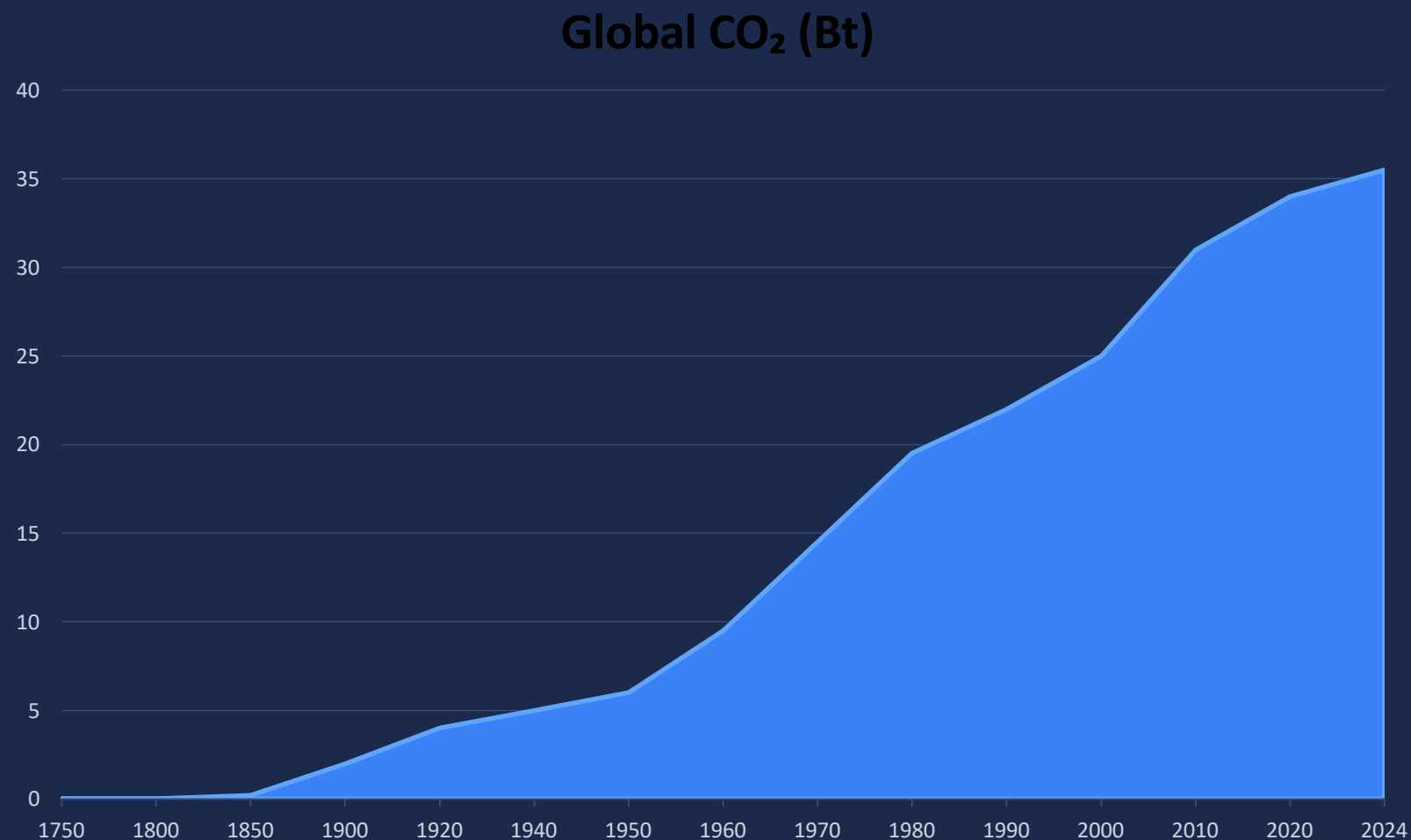
Fossil Fuels & Industry • 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

Authors: Hannah Ritchie and Max Roser

For policy analysts, climate negotiators, and sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Pre-1900

Negligible emissions
before industrialization

1950

~6 Bt/yr
Post-war acceleration begins

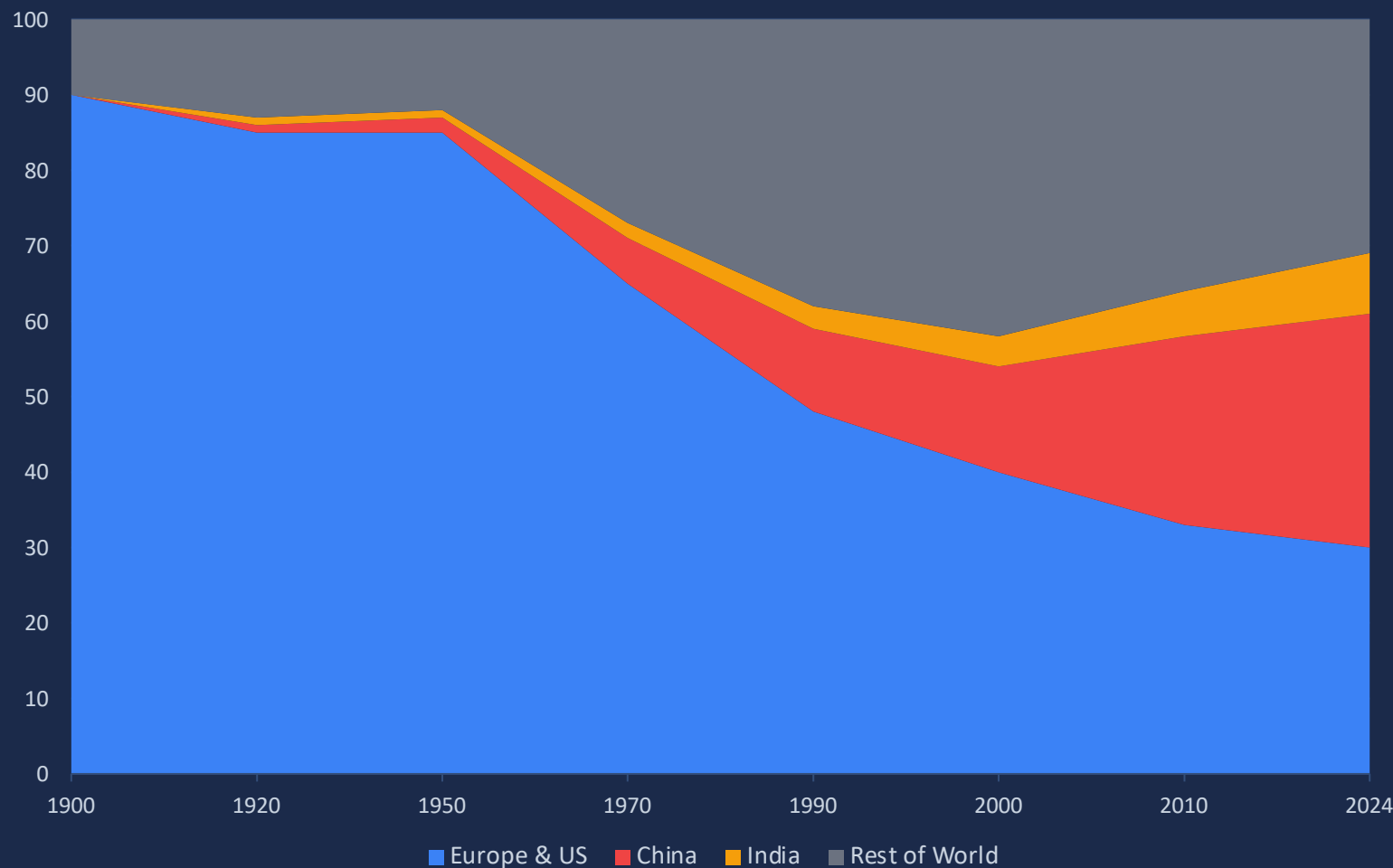
1990

~22 Bt/yr
Nearly quadrupled since 1950

2024

~35+ Bt/yr
Growth slowing but no peak

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



THE SHIFT

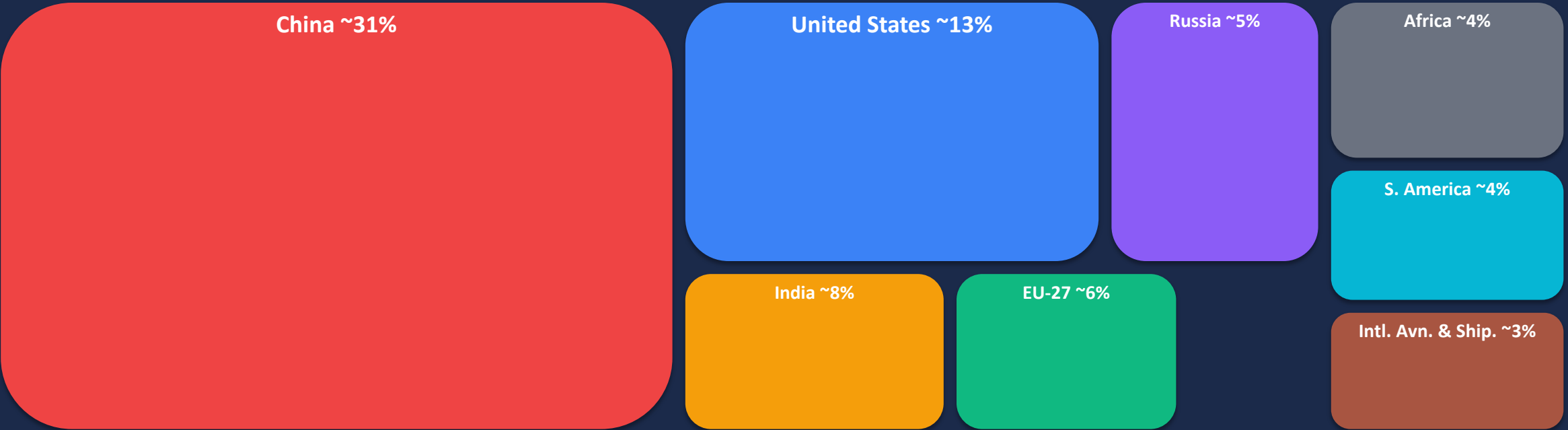
1900: Europe & US > 90%

2024: Europe & US < 30%

Asia now ~50% of global total

China's share rose from near zero in 1950 to ~31% today — the single largest shift in emission geography.

China Now Emits More Than One-Quarter of Global CO₂

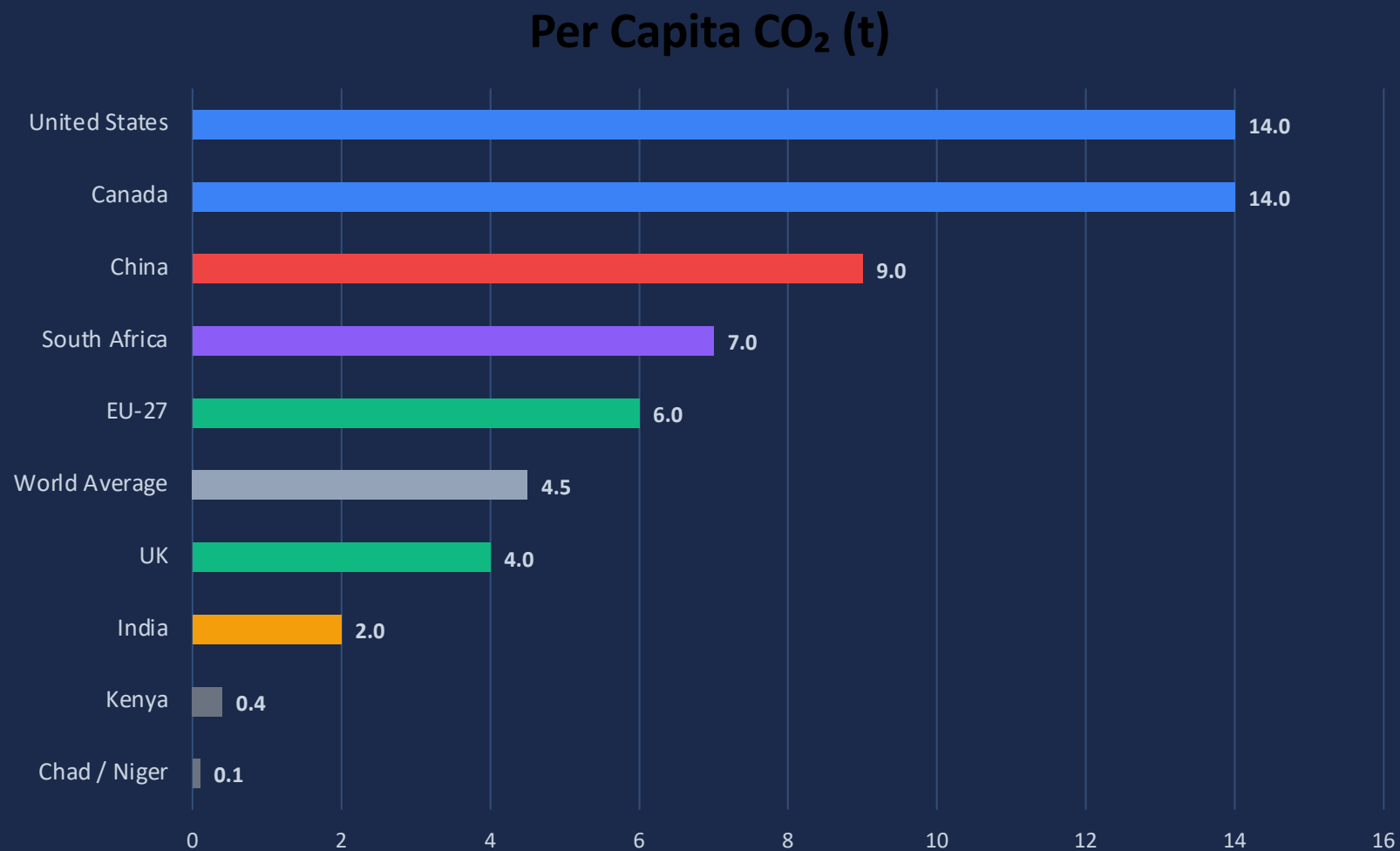


2024 Emission Shares (Approximate)

Country/Region	Share	Trend
China	~31%	Rising, growth slowing
United States	~13%	Declining
India	~8%	Rising
EU-27	~6%	Declining
Russia	~5%	Roughly stable
Africa (total)	~3–4%	Slowly rising

Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



150×

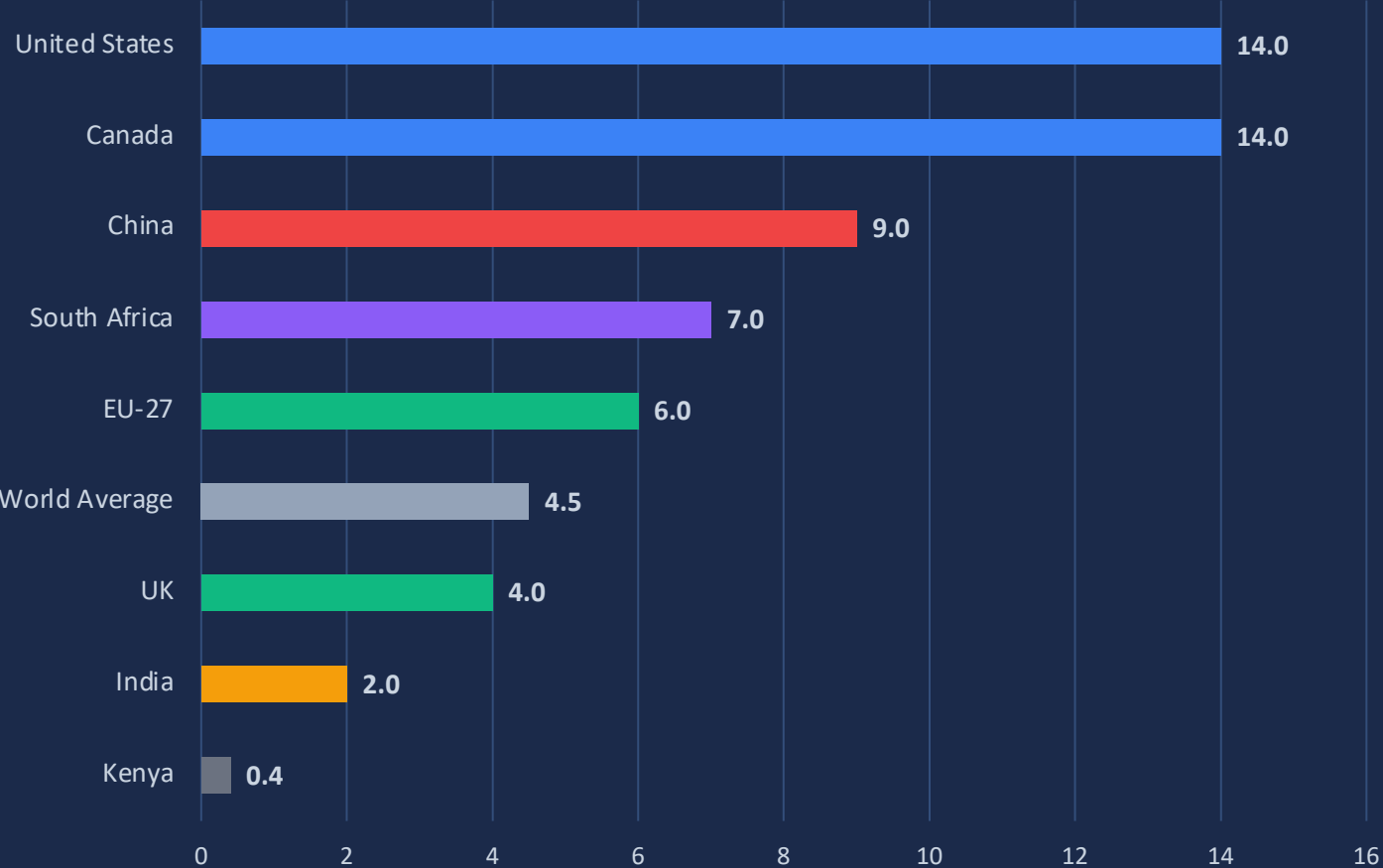
gap between the lowest and highest large-country per capita emitters. An average American emits in ~2 days what a person in Chad emits in a year.

ENERGY MIX MATTERS

France & UK emit far less per capita than Germany or the US despite similar living standards — nuclear and renewables drive the gap.

Per Capita Emissions: ~14 t in the US vs. ~9 t in China and ~2 t in India

Per Capita CO₂ (t)



LARGEST TOTAL ≠ LARGEST PER CAPITA

China emits ~31% of global CO₂ but only ~9 t per person — below the US at ~14 t.

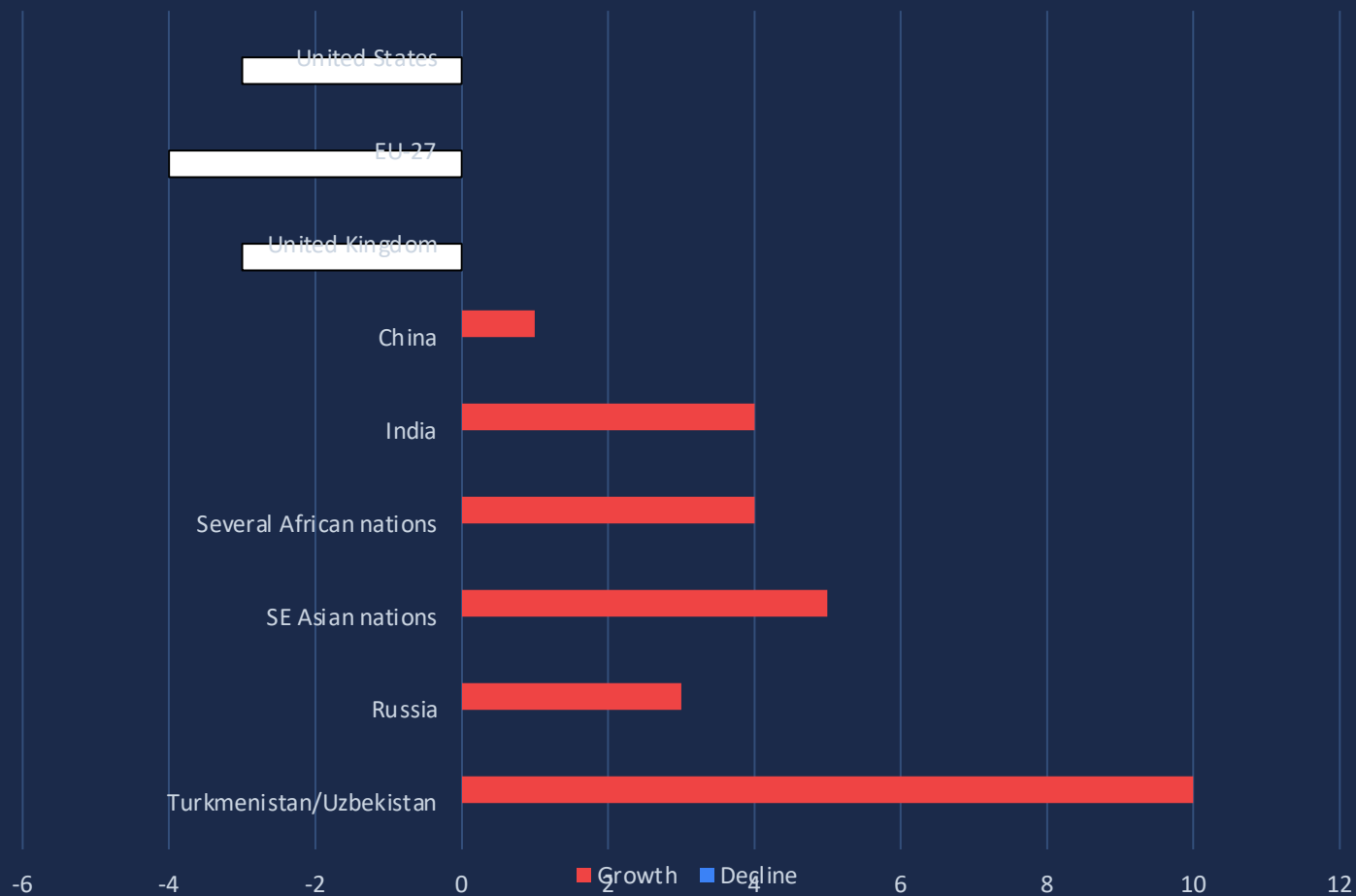
INDIA REMAINS VERY LOW

At ~2 t per person, India's per capita emissions are less than half the world average of ~4.5 t.

WEALTHY NATIONS DIVERGE

Oil-producing nations (Qatar, UAE) top the chart. Among large economies, the US/Canada (~14 t) dwarf the EU (~6 t) and UK (~4 t).

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



DECLINING

US, EU, and UK saw emission declines in 2024, continuing a multi-year trend in many developed economies.

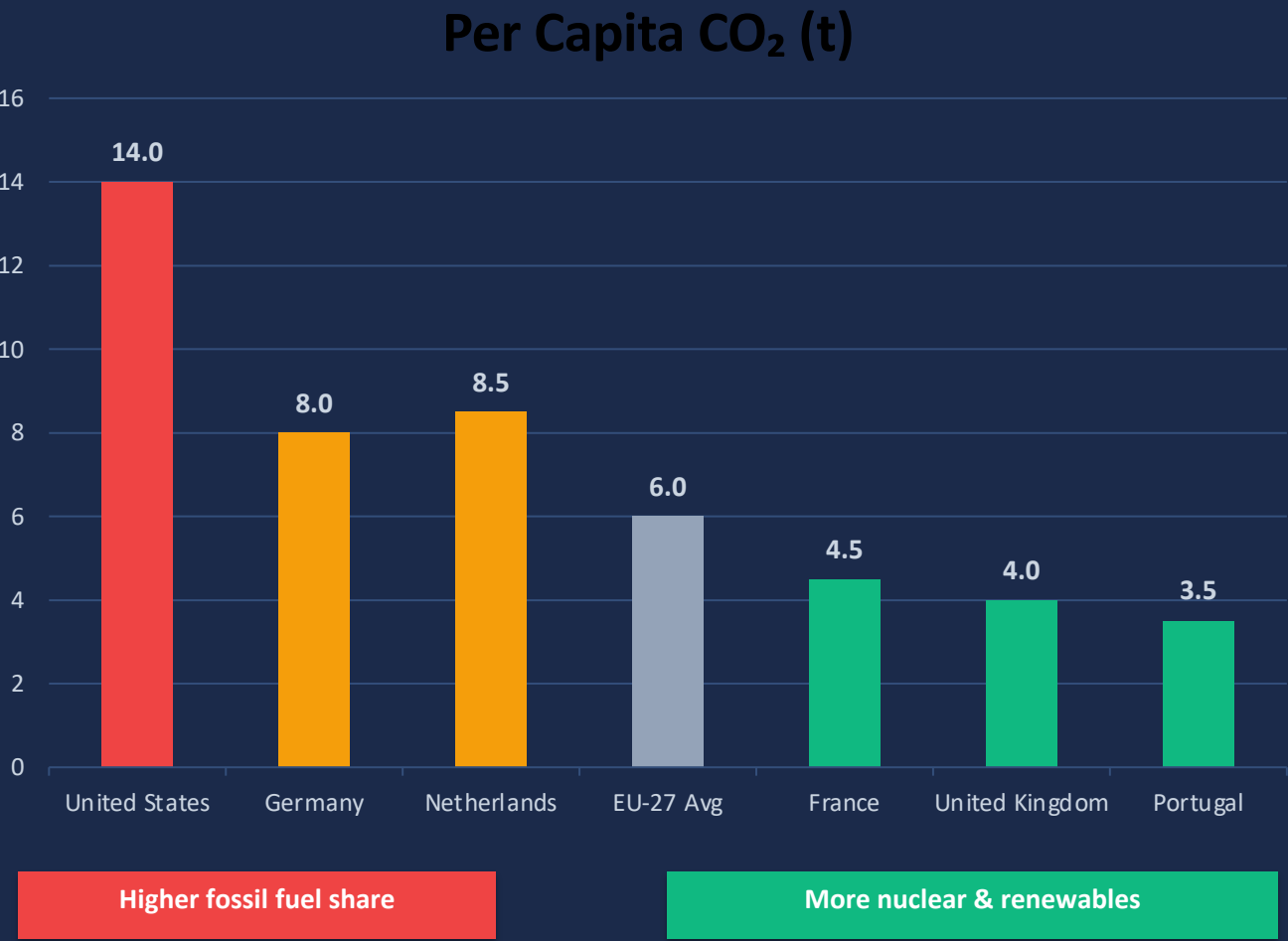
GROWING

Russia, parts of SE Asia, and several African nations experienced increases in 2024.

HOTSPOTS

Central Asian nations (Turkmenistan/Uzbekistan region) saw particularly large increases exceeding 10%.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



THE KEY INSIGHT

Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix.

France, the UK, and Portugal have much lower per capita emissions than Germany, the Netherlands, or the US.

The difference is largely because a higher share of their electricity comes from nuclear and renewable sources rather than fossil fuels.

This demonstrates that energy policy and technology choices can decouple prosperity from emissions.

Key Takeaways: Responsibility Is Shared but Unequal

01

NO PEAK YET

Global CO₂ emissions from fossil fuels exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth.

02

DRAMATIC SHIFT IN RESPONSIBILITY

Europe and the US dominated emissions historically (>90% in 1900) but now account for less than one-third; China alone emits ~31% of the global total.

03

MASSIVE PER CAPITA INEQUALITY

A 150× gap exists between the lowest emitters (Sub-Saharan Africa, ~0.1 t/person) and the highest large-country emitters (US/Canada, ~14 t/person).

04

CHINA: LARGEST EMITTER, NOT LARGEST PER CAPITA

China is the largest annual emitter but per capita (~9 t) still well below the US (~14 t).

05

ENERGY POLICY MATTERS

Countries with similar prosperity levels can have very different per capita emissions — energy mix choices can decouple prosperity from emissions.

Data: Global Carbon Budget 2025 via Our World in Data | Authors: Hannah Ritchie and Max Roser